EXERCISES

1. Given a request to an API at `https://api.example.com:8443/test/1`, which of the following URIs would be running on the same origin according to the same-origin policy?

a) `http://api.example.com/test/1`

b) `https://api.example.com/test/2`

c) `http://api.example.com:8443/test/2`

d) `https://api.example.com:8443/test/2`

e) `https://www.example.com:8443/test/2`

4.2 Token-based authentication

Let’s suppose that your users are complaining about the drawbacks of HTTP Basic authentication in your API and want a better authentication experience. The CPU overhead of all this password hashing on every request is killing performance and driving up energy costs too. What you want is a way for users to login once and then be trusted for the next hour or so while they use the API. This is the purpose of token-based authentication, and in the form of session cookies has been a backbone of web development since the start. When a user logs in by presenting their username and password, the API will generate a random string (the token) and give it to the client. The client then